

## Exercise for Lecture Virtual Reality and 3D Interaction

Dr. Nico Feld

- Cross Reality -

01.06.2026

### Submission

1. Fork the branch **04** of the exercise repository:

[https://gitlab.rlp.net/hci-trier-public/vr/2026\\_exercises](https://gitlab.rlp.net/hci-trier-public/vr/2026_exercises)

2. Do this exercise on your forked branch.
3. **To Submit:** Create a merge request to the exercise repository on branch **04**.
4. The submission should **at least** include the following files/folders:
  - Assets
  - ProjectSettings
  - Packages
  - readme.txt

**If any of these files/folders are missing, you will not pass this exercise!**

### Task 1 - Identification of a Workflow

For the final project, your task is to extend an existing workflow with cross-reality and implement a demonstrator containing all main functionalities.

Your chosen existing workflow should meet the following requirements:

- The workflow must include a digital or analog device.
- The workflow must not be entertainment (like a game), but work-focused (like office work).

Explain your decision and the core components of your workflow in detail! (e.g., which interactions does the workflow consist of?)

**Example Workflow 1:** *Creating 3D CAD Models on a PC for machinery for a factory with Blender.*

*Core components: Manipulating the Model, View change, Export/Import, ...*

**Example Workflow 2:** *Operating a mechanical signal box.*

*Core components: Physical interaction with the signal boxes, interaction is recognized by the system, switches are turning, ...*

### Task 2 - Decision for a Cross-Reality System

Once you have found a suitable workflow, consider how it could be extended via Cross-Reality. To do this, first decide which type of Cross-Reality System you want to implement. As explained in the lecture, you can choose between:

1. Transitional CR-Systems
2. Substitutional CR-Systems
3. Multi-User CR-Systems

Explain your decision!

**Example Workflow 1:** *Substitutional: The user is immersed in a virtual replica of the factory, but can see and interact with the physical desk and PC to model with Blender.*

**Example Workflow 2:** *Transitional: The user can operate the mechanical signal box in reality, but can switch to VR to get an overview of the tracks.*

Explain, from a design perspective, in detail the core components of your chosen workflow you aim to enhance with Cross-Reality and why. Also, explain, from a design perspective, which core components you don't aim to enhance with Cross-Reality and why.

### **Task 3 - Implementation of a Demonstrator**

Lastly, implement a demonstrator that includes all core components from above on an exemplary level. This means that every core component you identified must work on a basic level. Not every core component must work flawlessly, but every core component must be implemented.

**Example Workflow 1:** *The user is immersed in VR, the user can see and interact with the PC, the virtual environment, and Blender are somehow linked, ...*

**Example Workflow 2:** *The user has some extent of a physical signal box, the interaction with the signal boxes is registered by the system, the user can switch to VR through a transition, ...*

### **Interim Presentation on the 18.06.**

For the interim presentation, you must prepare Tasks 1 and 2 as a 10-minute presentation. Please include images and/or videos in your presentation! You must incorporate the feedback you get back into Tasks 1-3.

### **Final Presentation on the 16.07.**

For the final presentation, you must prepare your final Tasks 1-3 as a 10-minute presentation. Please include images and videos of your demonstrator in your presentation! Further, you must have a running version of your demonstrator prepared as a demo.

### **Access to the HCI-Makers Space**

During this exercise, you can get access to the HCI-Makers Space, e.g., to create custom input devices. There, you have access to a 3D printer, a laser cutter, and various electronics. You are encouraged to use this offer, but are not required to. If you are interested, just contact me: [feldn@uni-trier.de](mailto:feldn@uni-trier.de)

## General Information & Submission

- **Deadline:** 15.07.2026, 23:59.
- **Presentation:** 16.07.2026, 10:00. (Around 10 min each)
- **Version:** We will only use Unity version 6000.4.2f1:  
<https://unity.com/releases/editor/archive>
- **Dokumentation:** Include a *readme.txt* file with your submission that contains your thoughts and design choices for each decision/sub-task. This is a major part of the grading! Further, the file should document how your program works (key bindings, etc.). **Add your Student ID Number (Matrikelnummer) to the readme.txt!**
- **External Resources:** You may use external resources (scripts, models, etc.), as long as you cite them in the *readme.txt* and they do not provide a trivial solution to the problem at hand. If you have any questions, please ask.
- **Submission:** To submit, create a merge request to the exercises repository. Please delete any unnecessary assets from your Asset folder beforehand. Your submission should also include an APK that can run on the Quest 3. Make sure your repository is publicly accessible.
- **Your submission will only be graded if the APK is executable and can also be run in the editor. Also, make sure your program does not generate any runtime errors.**